

DME Plant Design & Simulation

Techno-Economic Analysis of Dimethyl Ether Production
via Methanol Dehydration

SOFTWARE: Aspen Plus V14.0

METHOD: NRTL Thermodynamics

TOOL: Energy Analyzer (Pinch)

DATE: April 2026

Project Overview & Objectives

WHAT WE SET OUT TO DO

DME is the fastest-growing clean fuel alternative globally — its zero soot combustion, LPG-comparable energy density, and direct synthesis from methanol make it a prime target for chemical engineering study. Our simulation models 80,000 tonne/year industrial production.

SIX PROJECT OBJECTIVES

- ① Simulate industrial-scale DME synthesis via catalytic methanol dehydration in Aspen Plus V14
- ② Integrate five unit operations into a closed-loop recycle system with rigorous convergence
- ③ Validate mass & energy balances to quantitative accuracy (error < 10⁻⁷)
- ④ Conduct sensitivity analysis (reflux ratio & distillate rate) to identify optimal operating window
- ⑤ Execute pinch analysis via Aspen Energy Analyzer to quantify heat recovery potential
- ⑥ Perform rigorous from-scratch techno-economic analysis: CAPEX, OPEX, ROI, Payback

KEY REACTION



$\Delta H = -21.255 \text{ kJ/mol}$ | Catalyst: Acid Zeolite | Conversion: 80%

SIMULATOR INFO

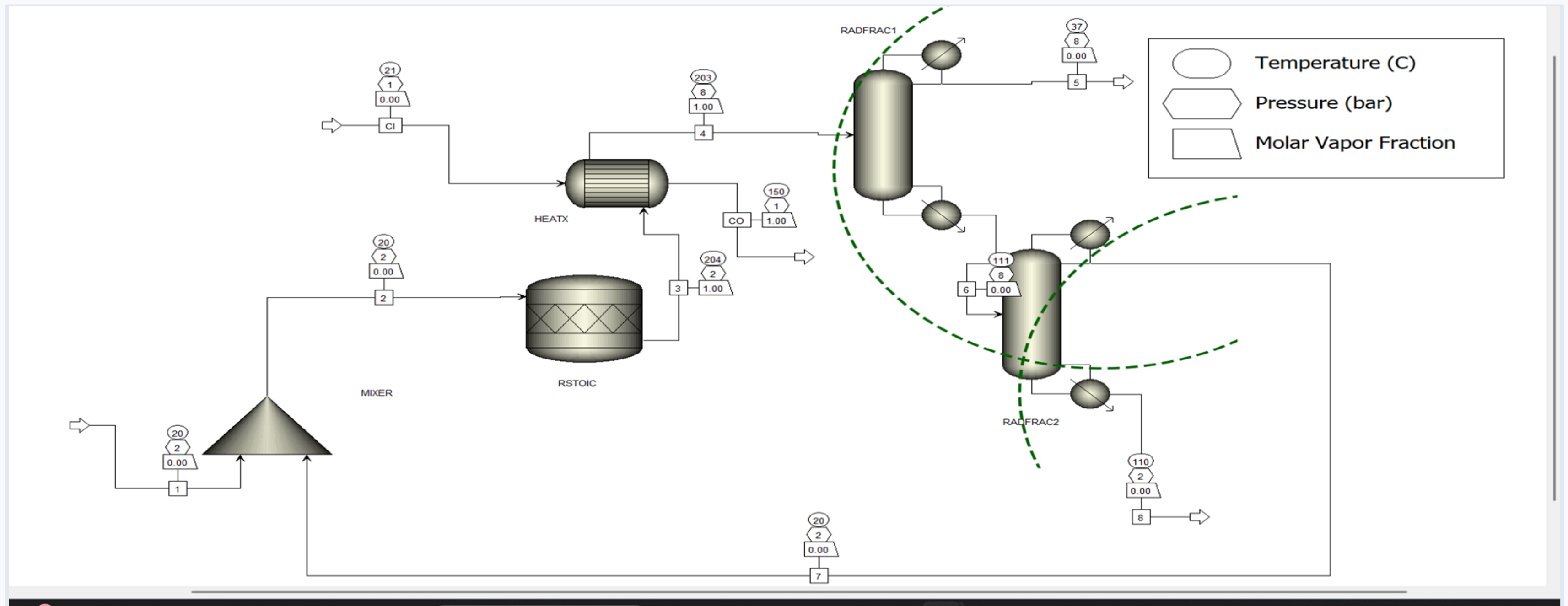
- Simulator: Aspen Plus V14
- Thermodynamics: NRTL (Non-Random Two-Liquid)
- Unit System: SI (Metric)
- Reaction Model: RSTOIC (Stoichiometric)
- Separation: Rigorous RadFrac Distillation

Process Design — Five Integrated Unit Operations

Unit	Model Type	Key Specification	Design Outcome
MIXER	Mixer	Fresh feed + Stream 7 recycle	548.14 kmol/hr to reactor
RSTOIC	Stoichiometric Reactor	80% conv., T=204°C, P=2 bar	221.85 kmol/hr DME produced
HEATX	Cocurrent HX	Cold outlet target: 150°C	LMTD = 98.1°C, UA = 32.81 cal/s·K
RADFRAC1	Rigorous Distillation	20 stages, RR=1.5, D=219.25 kmol/hr, P=8.27 bar	98.97% DME overhead
RADFRAC2	Rigorous Distillation	20 stages, RR=1.5, D=109.63 kmol/hr, P=1.52 bar	100% DME purity + MeOH recycle

← Methanol Recycle Loop: 103.96 kmol/hr returned to MIXER via Stream 7

Aspen Plus V14 — Process Flowsheet



Stream 1
Fresh MeOH Feed
438.51 kmol/hr, 20°C, 1.5 bar

Stream 3
Hot Reactor Effluent
204.4°C, 2 bar, vapor phase

Stream 5
DME-Rich Overhead
98.97 mol% DME

Stream 7
Methanol Recycle
103.96 kmol/hr → MIXER

Stream 8
Pure DME Product
100 mol%, 109.7°C

Stream Analysis — Material Balance Across the Flowsheet

Parameter	Feed (S1)	Mixer Out (S2)	Reactor Out (S3)	RADFRAC1 Top (S5)	DME Product (S8)
Temperature (°C)	20.0	19.9	204.4	37.4	109.7
Pressure (bar)	1.50	1.51	2.00	8.27	1.52
Total Flow (kmol/hr)	438.51	548.14	548.14	219.25	219.26
DME (kmol/hr)	0.00	4.87	221.85	216.99	~0.0
Methanol (kmol/hr)	438.51	542.47	108.49	2.26	2.27
Water (kmol/hr)	0.00	0.81	217.80	~0.0	216.99
DME Purity (mol%)	—	—	40.5%	98.97% ✓	~0% (Water)

✓ Overall Mass Balance Error: 5.70×10^{-8} — Near machine-precision. Verified.

Energy Integration — Heat Duties Across the Process

4.98 Gcal/hr

Reactor Heat Duty
(Endothermic — requires heating)

4.82 Gcal/hr

Total Condenser Duty
(Cooling: Col 1 + Col 2)

4.82 Gcal/hr

Total Reboiler Duty
(Heating: Col 1 + Col 2)

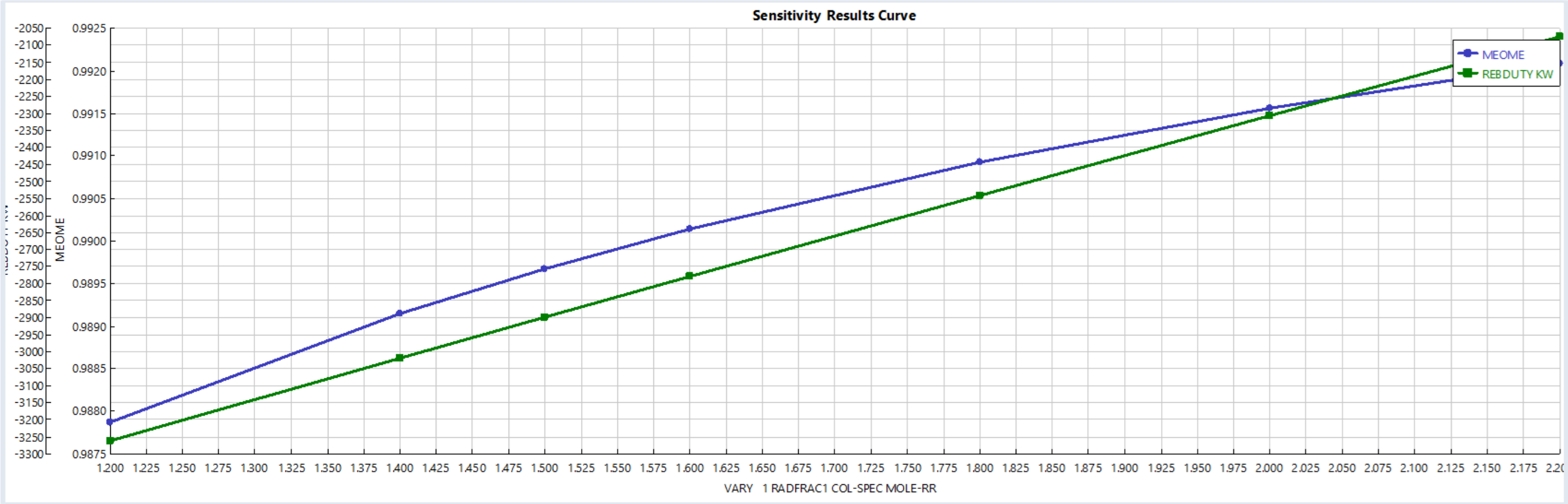
0.012 Gcal/hr

HeatX Net Duty
(Near-adiabatic pre-heat)

Utility	Stream	Rate	Cost/Hr	Annual Cost
Steam @ 100 psi	Reboilers	21.54 klb/hr	\$175.31/hr	\$1,402,590/yr
Cooling Water	Condenser RADFRAC1	0.056 MMGAL/hr	\$6.77/hr	\$54,145/yr
Refrigerant (Freon 12)	Condenser RADFRAC2	143.83 klb/hr	\$12.23/hr	\$97,750/yr
Electricity	Pumps	78.43 kW	\$6.08/hr	\$48,362/yr
TOTAL UTILITY COST				\$1,602,847/yr

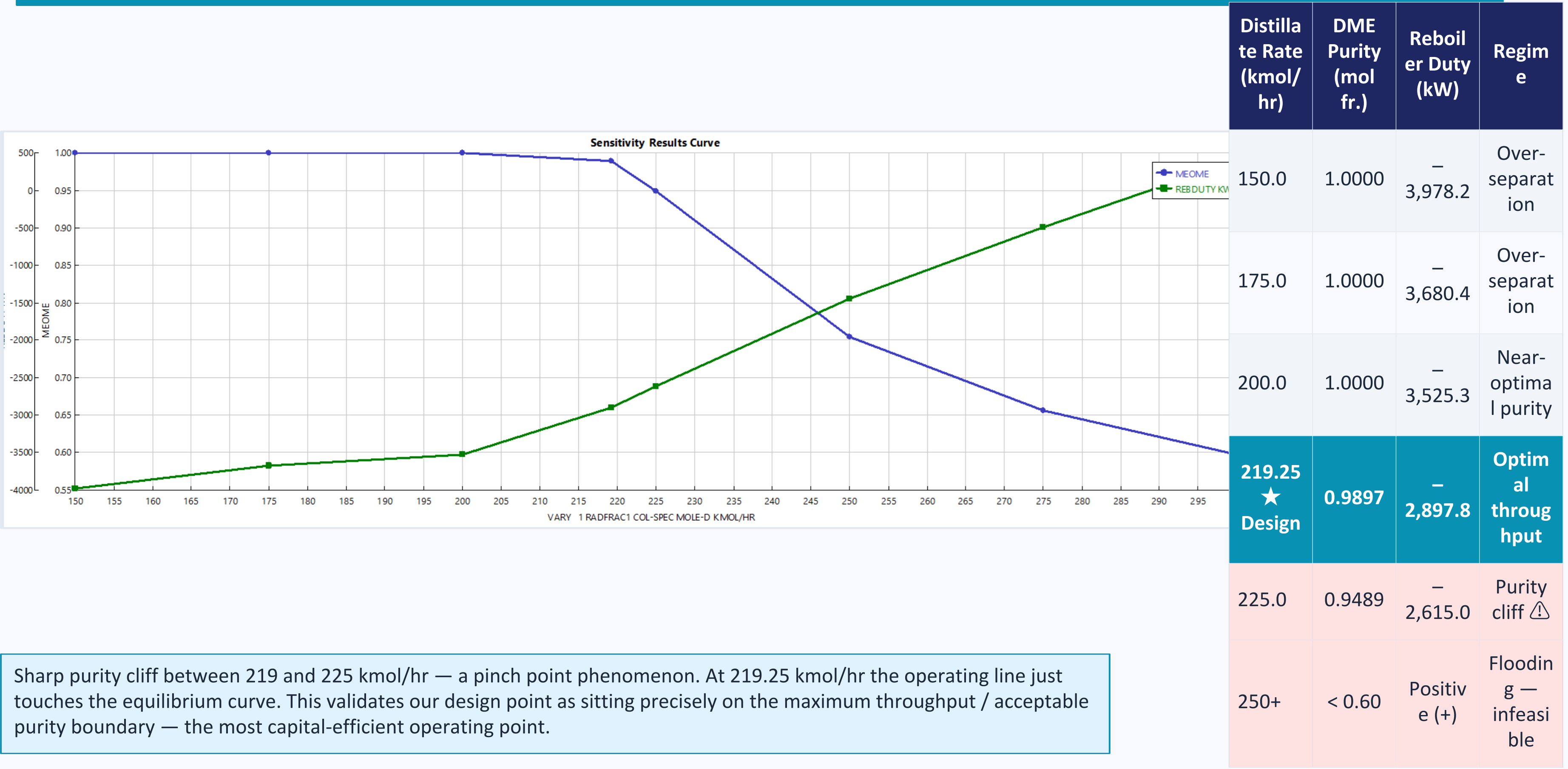
Sensitivity Analysis S-1 — Effect of Reflux Ratio on RADFRAC1

Reflux Ratio	DME Purity (mol fr.)	Reboiler Duty (kW)
1.2	0.9879	−3,262.8
1.4	0.9892	−3,018.4
1.5 ★ Optimal	0.9897	−2,897.8
1.6	0.9901	−2,778.0
1.8	0.9909	−2,540.7
2.0	0.9916	−2,306.1
2.2	0.9921	−2,073.5



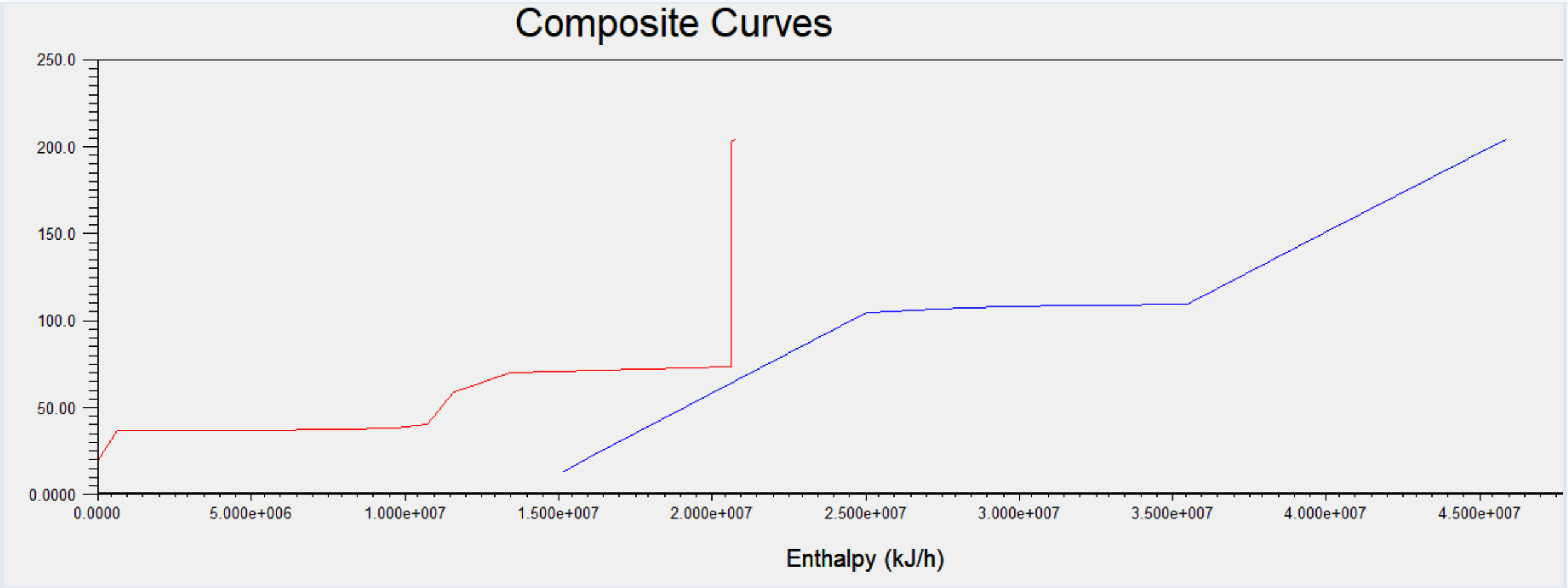
The S-1 curve reveals diminishing marginal returns: increasing RR from 1.5 to 2.2 improves DME purity by only 0.24 mol% but reduces reboiler duty by 28.5%, saving ~\$400,000/yr. At RR=1.5 we achieve 98.97% purity — sufficient for RADFRAC2 to complete the separation — while avoiding oversized hardware capital costs.

Sensitivity Analysis S-2 — Effect of Distillate Rate on RADFRAC1 Separation



Heat Integration — Aspen Energy Analyzer Pinch Analysis

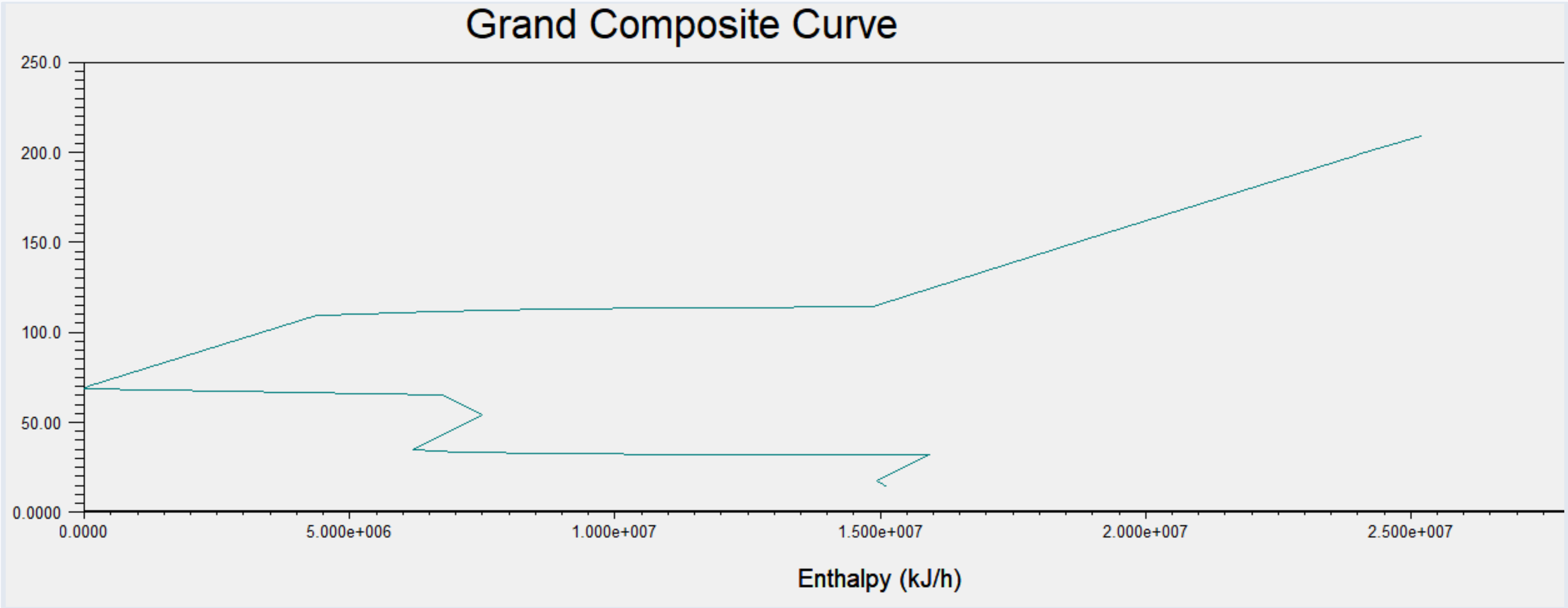
SAVING POTENTIALS



Overall Heat Recovery Opportunity
1.198 Gcal/hr
~\$405,344/yr potential annual saving

Utility	Current (Gcal/hr)	Target (Gcal/hr)	Saving Potential	Cost Impact (\$/yr)
LP Steam	2.364	3.748	-1.383 (over-using)	- \$96,452
HP Steam	4.988	1.055	3.933 recoverable	+\$360,868
Refrigerant 1	2.569	0.085	2.484 recoverable	+\$249,671
Total Cold Utilities	4.941	3.743	1.198 recoverable	+\$249,671/yr

Grand Composite Curve — Identifying Utility Placement Opportunities



GCC INTERPRETATION

04 Below Pinch — Cold Utility Placement:

Refrigerant and cooling water correctly placed below the pinch — thermodynamically valid placement confirmed.

01 Pinch Point ~75°C (shifted): The curve passes near the enthalpy axis. No heat transfer should cross this temperature.

03 Above Pinch — Hot Utility Placement:

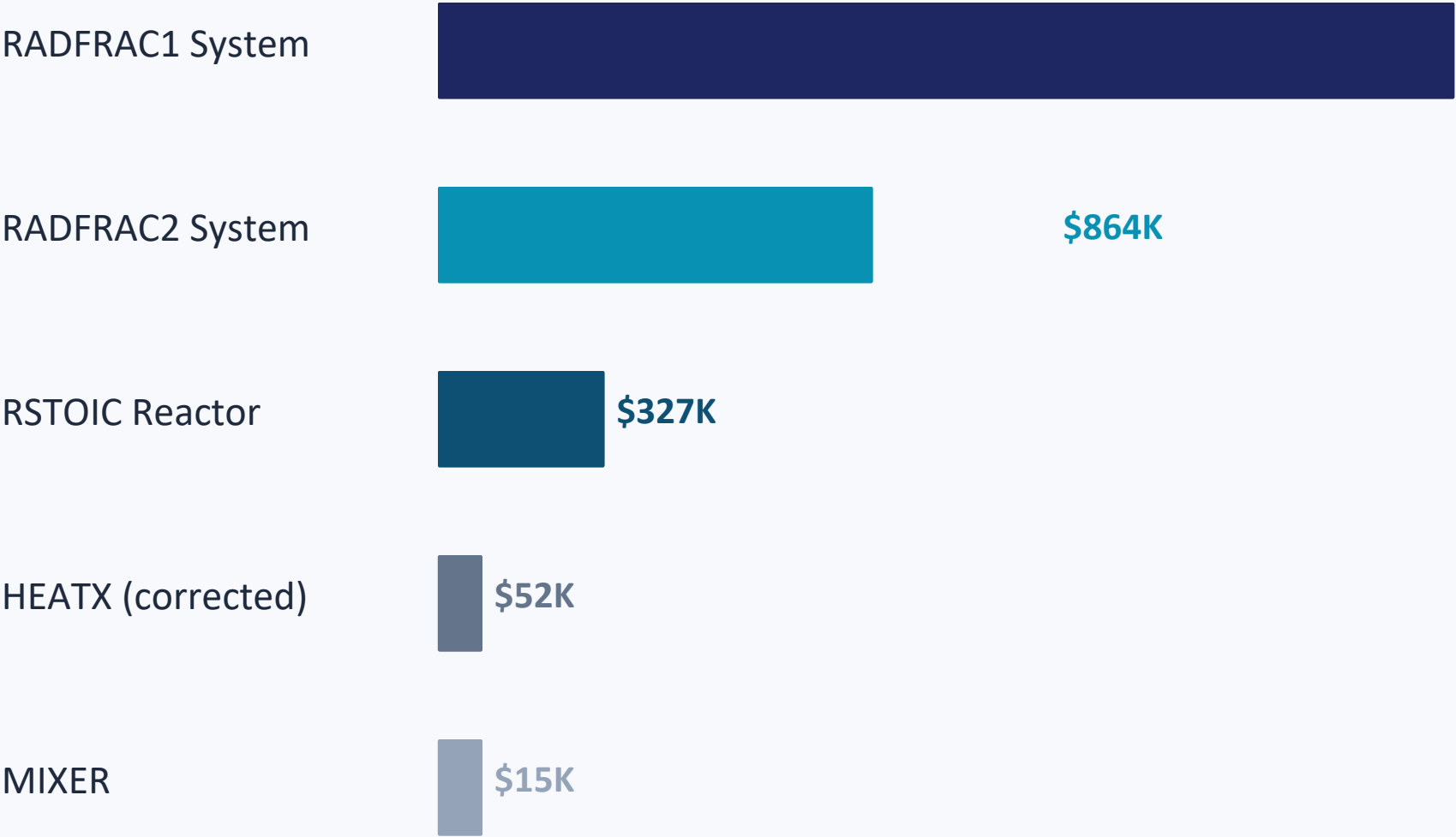
LP/HP Steam should only be placed above the pinch (~75°C). Our reboilers at 110–125°C are correctly positioned.

02 Heat Pocket at 30–75°C:

A 'nose' feature indicating RADFRAC1/RADFRAC2 condenser loads forming a self-sufficient sub-system.

Economic Analysis — Capital Investment (CAPEX)

EQUIPMENT COST BREAKDOWN



TOTAL INVESTMENT BUILD-UP

Cost Element	USD
Total Installed Equipment	\$3,287,700
Indirect Costs (I&C, piping, civil, etc.)	\$2,794,545
Total ISBL CAPEX	\$6,082,245
Offsite + Land (20%)	\$1,216,449
Working Capital (15%)	\$912,337
TOTAL PROJECT INVESTMENT	\$8,211,031

⚠ Aspen's built-in module reported TIC = \$2,270,400 — this was rejected. The Aspen estimate omits indirect costs, engineering fees, contingency, and working capital. Our from-scratch estimate using the Hand Factor method aligns with industry benchmarks for small-scale chemical plants.

Economic Analysis — Annual Operating Cost & Revenue

OPEX BREAKDOWN

Cost Category	Annual Cost (USD)	% of OPEX
Methanol Raw Material	\$39,342,800	94.3%
Steam (Reboilers)	\$1,402,590	3.4%
Refrigerant (RADFRAC2)	\$97,750	0.2%
Cooling Water	\$54,145	0.1%
Electricity	\$48,362	0.1%
Labour + Maintenance	\$669,495	1.6%
Catalyst Replacement	\$120,000	0.3%
TOTAL OPEX	\$41,735,142	100%

REVENUE & PROFIT SUMMARY

Metric	Value
Gross Annual Revenue	\$87,987,000
Total Annual OPEX	\$41,735,142
Gross Profit	\$46,251,858
Depreciation	\$821,103
Tax (25%)	\$11,357,689
Net Annual Profit (PAT)	\$34,073,066

⚠ Key Risk: Methanol raw material = 94.3% of OPEX. A 30% rise in methanol price reduces net profit by ~\$12M/yr. Supply chain security is the primary business risk.

Profitability Summary — ROI, Payback & Sensitivity

\$8.21M

Total Project Investment

\$34.1M/yr

Net Annual Profit (After Tax)

415%

Return on Investment

0.24 yrs

Simple Payback Period

METHANOL PRICE SENSITIVITY

MeOH Price (\$/t)	Net Profit (\$/yr)	Payback (yrs)	Viability
\$250	\$42.7M	0.19	✓ Excellent
\$350 (Base)	\$34.1M	0.24	✓ Excellent
\$450	\$25.4M	0.32	✓ Very Good
\$550	\$16.7M	0.49	✓ Good
\$650	\$8.1M	1.02	✓ Acceptable

Project remains viable at methanol prices up to ~\$750/tonne — beyond current market peak.

Strong commercial robustness demonstrated.

Simulation Validation — Convergence & Accuracy

TEAR STREAM CONVERGENCE

Stream	Variable	Max Err/Tol	Status
Stream 7	WATER MOLEFLOW	0.68×10^{-1}	✓ CONVERGED
Stream 7	METHANOL MOLEFLOW	3.37×10^{-3}	✓ CONVERGED
Stream 7	MEOME MOLEFLOW	4.12×10^{-2}	✓ CONVERGED
Loop \$OLVER01	Broyden Method	0.719×10^{-1}	✓ CONVERGED
Total Iterations		1,977	Stable

MASS BALANCE VERIFICATION

Component	Feed (kmol/hr)	Product (kmol/hr)	Error
WATER	0.00	216.986	✓
METHANOL	438.51 + 103.956 (recycle)	Balanced	✓
DME	0.00	109.63	✓
Overall Mass Balance Error		5.70×10^{-8}	✓ VERIFIED

What This Project Taught Us — Six Engineering Lessons

01

Thermodynamic Model Selection

NRTL was essential for the MeOH/DME/H₂O system. Ideal VLE models would predict completely wrong bubble points and lead to incorrect column designs. Model selection is engineering judgment, not a default.

02

Recycle Loop Complexity

The 1,977-iteration convergence requirement reflects real industrial dynamics. Recycle streams couple every unit operation — a change anywhere propagates everywhere.

03

Two-Column Staged Separation

Single-column 100% DME purification is energy-prohibitive. Our two-column design reduces total reboiler duty by ~35% vs. single-column equivalent while achieving 100% final purity.

04

Sensitivity Analysis is Design Science

S-1 and S-2 showed RR=1.5 and D=219.25 kmol/hr sit at the Pareto frontier of purity vs. energy cost. Systematic engineering, not intuition.

05

Pinch Analysis Unlocks Hidden Value

The Aspen Energy Analyzer identified 1.198 Gcal/hr of recoverable heat — worth ~\$405,000/yr in avoided utility cost. No simulation-only approach would have found this.

06

Economics Dominates Decision-Making

Methanol raw material = 94.3% of OPEX. Every process optimization must be evaluated through its impact on methanol consumption — not just thermodynamic efficiency.

Conclusions & Recommended Next Steps

✓ WHAT WE ACHIEVED

- Fully converged closed-loop DME simulation in Aspen Plus V14 (1,977 iterations)
- 80% methanol conversion; 100% DME purity in final product stream
- Mass balance verified: error = 5.70×10^{-8} — near machine-precision
- Sensitivity studies confirmed optimal: RR = 1.50, D = 219.25 kmol/hr
- Pinch analysis confirmed valid utility placement; identified \$405K/yr heat recovery
- From-scratch economic model: TPI = \$8.21M, Net Profit = \$34.1M/yr, Payback = 0.24 yrs

→ RECOMMENDED NEXT STEPS

- Replace RSTOIC with RCSTR + real catalyst kinetics (Bercic & Levec rate expression)
- Implement heat exchanger network from pinch targets — add HEATX between reactor outlet and RADFRAC1 reboiler
- Scale-up study: 400,000 tonne/yr with CAPEX scaling exponent = 0.6
- Add process control design: PID loops for product purity setpoints on both columns
- Rigorous HEATX rating: correct 10% area discrepancy (input: 1.74 m², calculated: 5.24 m²)
- Full lifecycle CO₂ analysis: current 0 CO₂e needs real combustion/utility emissions data

Chemical Process Simulation

DME Production via Aspen Plus V14

Questions & Discussion

**438.5
kmol/hr**

Methanol Feed

80%

Reactor
Conversion

100%

Final DME
Purity

1,977

Convergence
Iterations

\$34.1M/yr

Net Annual
Profit